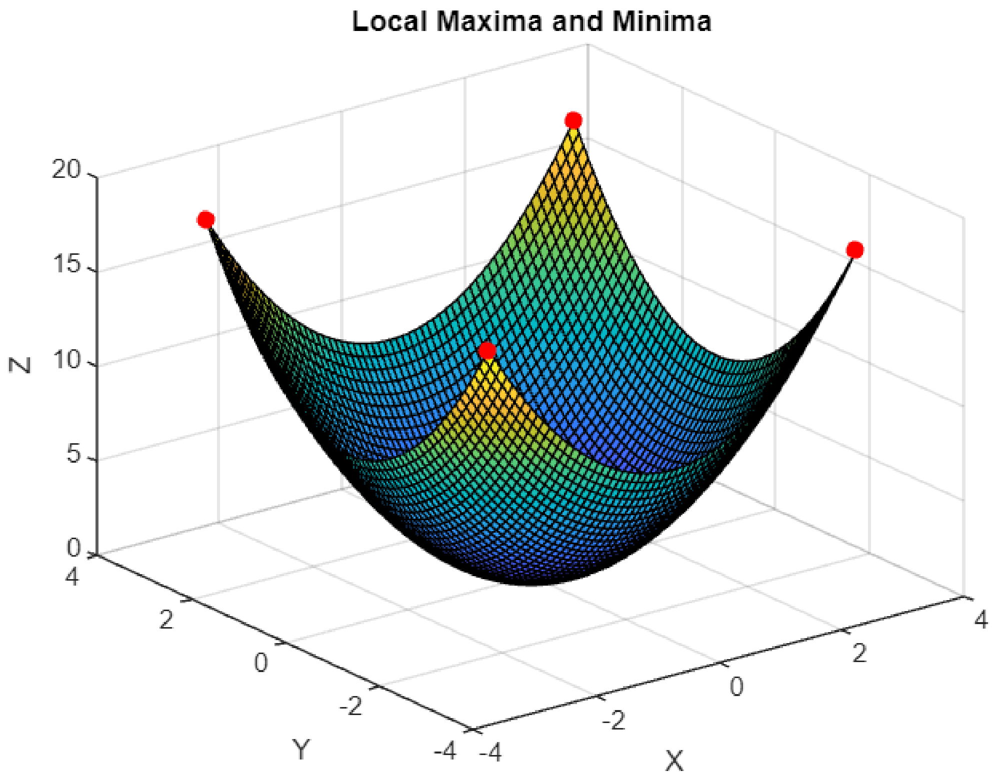
Amrita School of Engineering, Bengaluru-35
23MAT206 – Optimization Techniques
Lab Practice Sheet-2
Graphical solution of optimization problems up to 2 variables using MATLAB

MATLAB sample code provided for the following problems:

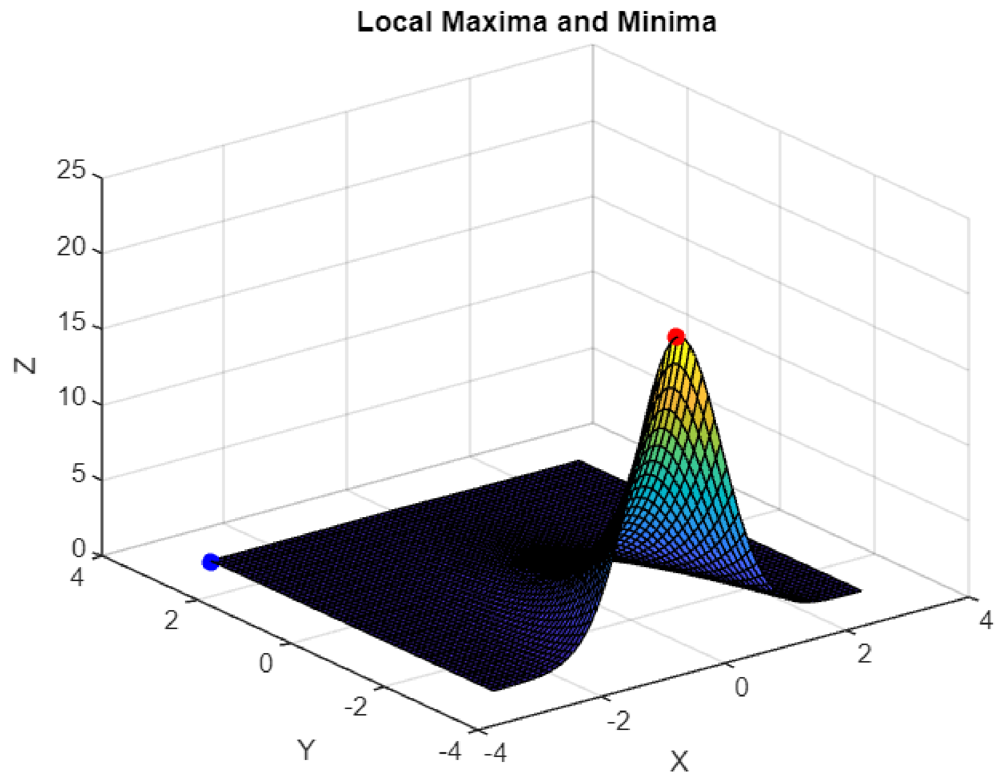
1. Find the local maxima and local minima of $x^2 + y^2 \quad x, y \in [-3,3]$.

```
x=-3:0.1:3;
y=x;
[x1,y1]=meshgrid(x,y);
z=x1.^2+y1.^2;
surf(x1,y1,z)
maximaMask=imregionalmax(z);
minimaMask=imregionalmin(z);
hold on;
plot3(x1(maximaMask),y1(maximaMask),z(maximaMask),'r.','MarkerSize',20);
plot3(x1(minimaMask),y1(minimaMask),z(minimaMask),'r.','MarkerSize',20);
hold off;
title("Local Maxima and Minima");
xlabel("X")
ylabel("Y")
zlabel("Z")
```



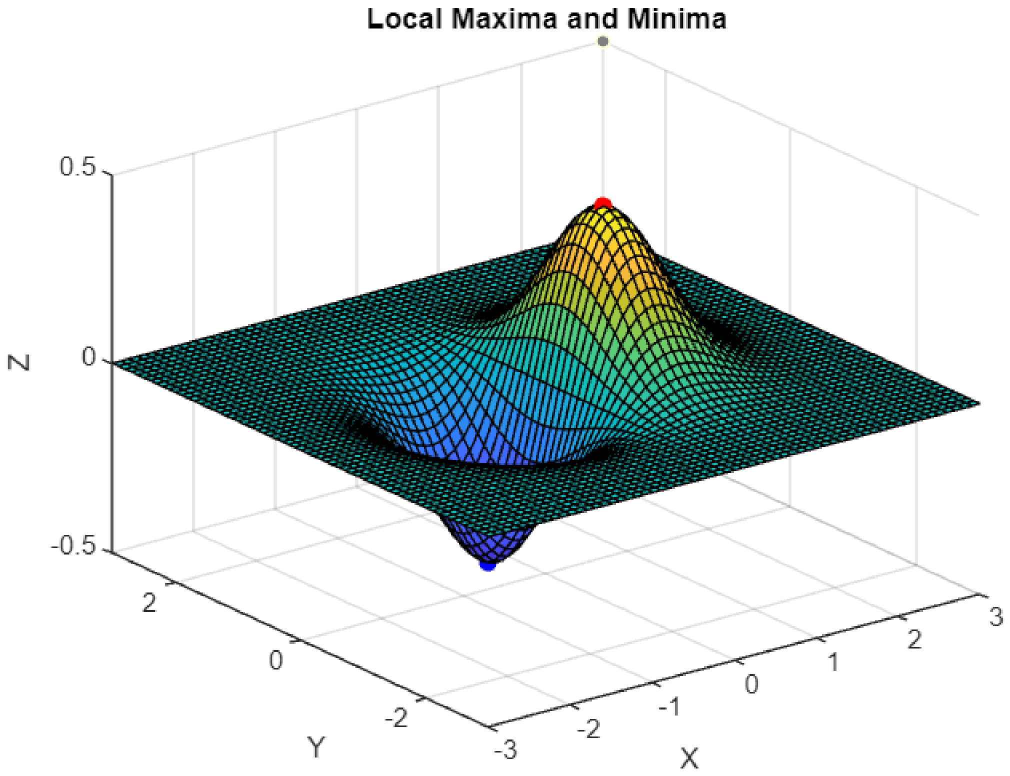
2. Find the local maxima and local minima of e^{-x^2+y} $x, y \, \epsilon[-3,3]$.

```
clc;
clear all;
x=-3:0.1:3;
y=x;
[x1,y1]=meshgrid(x,y);
z1=exp(-(x1.^2+y1));
[maxZ,maxIndex]=max(z1(:));
[minZ,minIndex]=min(z1(:));
[maxRow,maxCol]=ind2sub(size(z1),maxIndex);
[minRow,minCol]=ind2sub(size(z1),minIndex);
figure;
surf(x1,y1,z1);
hold on;
plot3(x1(maxRow,maxCol),y1(maxRow,maxCol),maxZ,'r.','MarkerSize',20);
plot3(x1(minRow,minCol),y1(minRow,minCol),minZ,'b.','MarkerSize',20);
hold off;
title("Local Maxima and Minima");
xlabel("X")
ylabel("Y")
zlabel("Z")
```



3. Find the local maxima and local minima of $xe^{-(x^2+y^2)}$ $x, y \epsilon[-3,3]$.

```
clc;
clear all;
x=-3:0.1:3;
y=x;
[x1,y1]=meshgrid(x,y);
z2=x.*exp(-(x1.^2+y1.^2));
[maxZ,maxIndex]=max(z2(:));
[minZ,minIndex]=min(z2(:));
[maxRow,maxCol]=ind2sub(size(z2),maxIndex);
[minRow,minCol]=ind2sub(size(z2),minIndex);
figure;
surf(x1,y1,z2);
hold on;
plot3(x1(maxRow,maxCol),y1(maxRow,maxCol),maxZ,'r.','MarkerSize',20);
plot3(x1(minRow,minCol),y1(minRow,minCol),minZ,'b.','MarkerSize',20);
hold off;
title("Local Maxima and Minima");
xlabel("X")
ylabel("Y")
zlabel("Z")
```

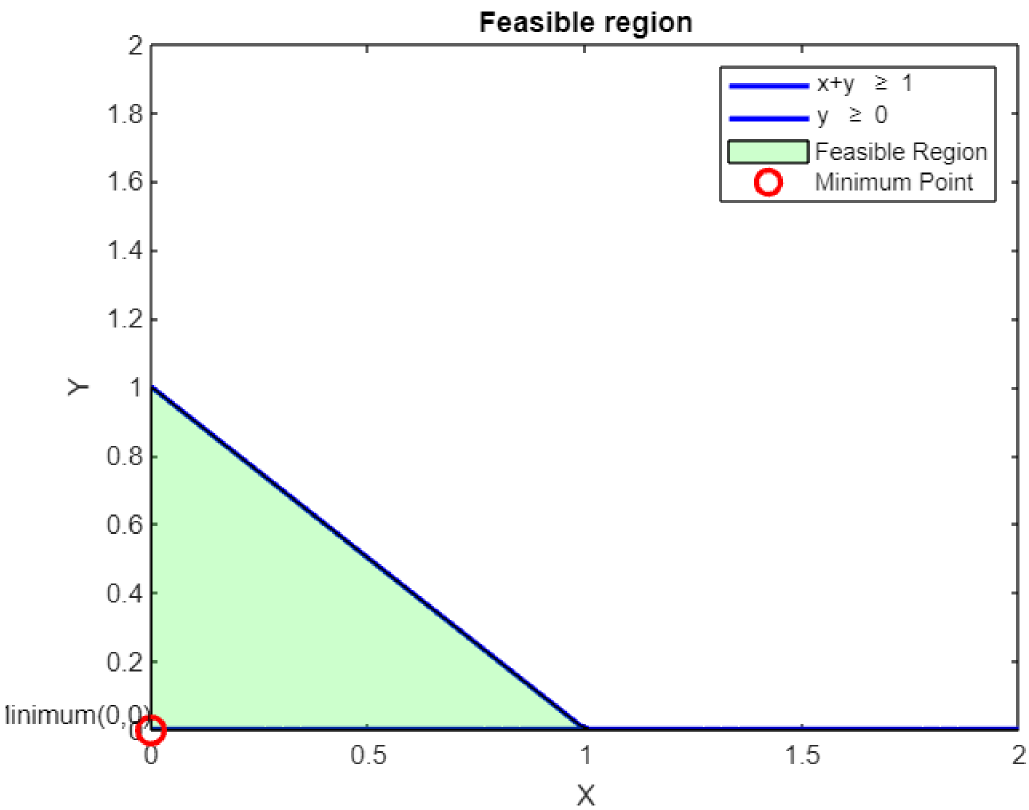


4. Minimize z= $x^2 + y^2$ subject to $x + y \geq 1; x, y \geq 0$

```
clc;
clear all;
x=linspace(0,2,100);
y1=max(0,1-x);
y2=zeros(size(x));
figure;
plot(x,y1,'b','LineWidth',2);
hold on;
plot(x,y2,'b','LineWidth',2);
xlabel("X")
ylabel("Y")
title("Feasible region");
ylim([0,2]);

fill([x,fliplr(x)],[y1,fliplr(y2)],'g','FaceAlpha',0.2);
objective=@(x,y) x.^2+y.^2;
[X,Y]=meshgrid(0:0.1:2,0:0.1:2);
Z=objective(X,Y);
Z((X+Y)>=1)= NaN;
[minZ,minIndex]=min(Z(:));
[minRow,minCol]=ind2sub(size(Z),minIndex);
minX=X(minRow,minCol);
minY=Y(minRow,minCol);
```

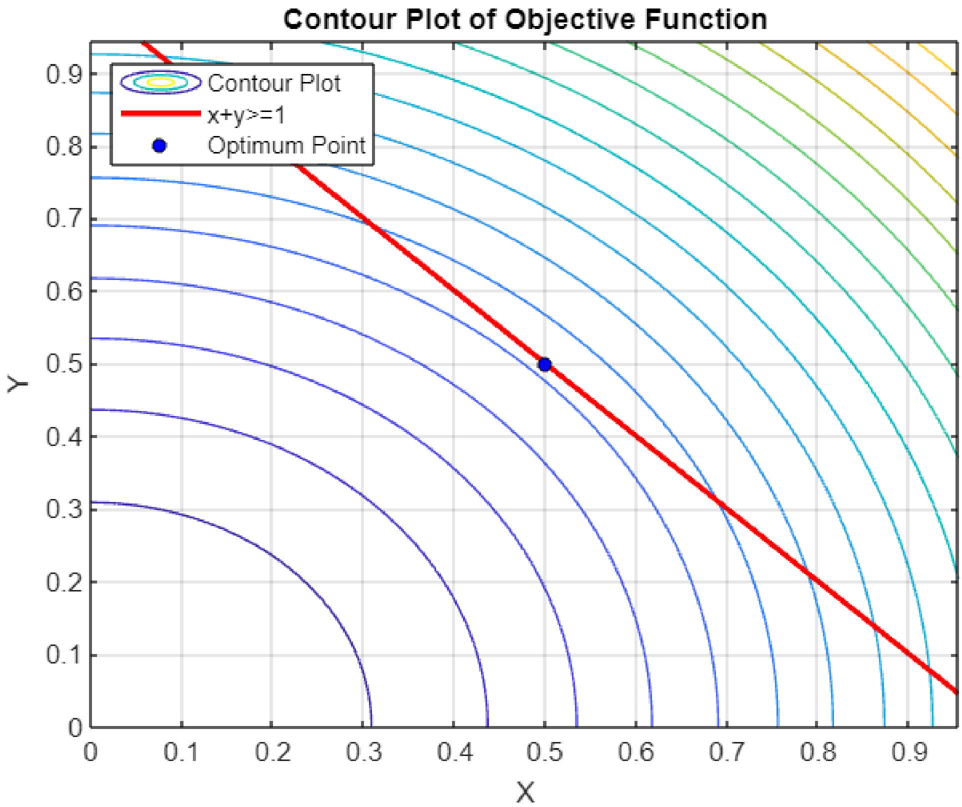
```
plot(minX,minY,'ro','MarkerSize',10,'LineWidth',2);
text(minX,minY,
['Minimum(',num2str(minX),',',num2str(minY),')'],'VerticalAlignment','bottom','Horiz
ontalAlignment','right');
xlabel("X")
ylabel("Y")
zlabel("Z")
legend('x+y \geq 1','y \geq 0','Feasible Region','Minimum Point');
hold off;
```



5. Minimize z= $x + y$ subject to $x^2 + y^2 \leq 1; x, y \geq 0$

```
clc;
fun=@(x) x(1)^2 + x(2)^2;
x0=[0,0];
A=[-1,-1];
B=-1;
lb=[0,0];
[x_opt,fval]=fmincon(fun,x0,A,B,[],[],lb,[]);
disp('Optimal Solution : ');
disp(['x = ',num2str(x_opt(1)),', y = ', num2str(x_opt(2))]);
disp(['Function value at optimum : ',num2str(fval)]);
```

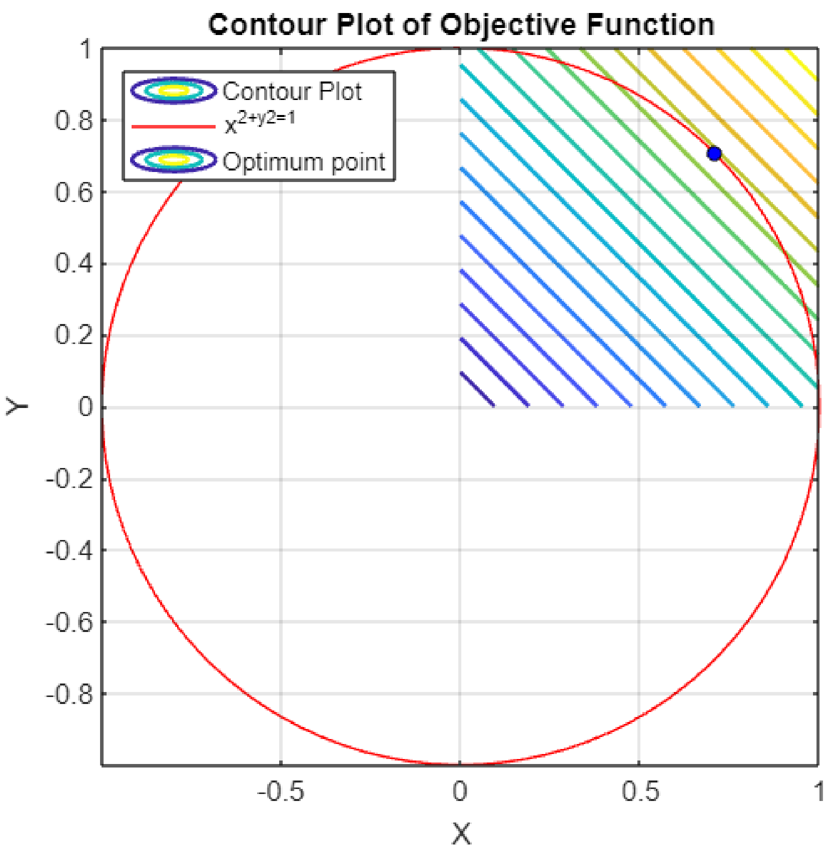
```
x=0:0.001:1;
y=0:0.001:1;
[X,Y]=meshgrid(x,y);
Z=X.^2+Y.^2;
contour(X,Y,Z,20);
hold on;
plot(x,1-x,'r','LineWidth',2);
plot(x_opt(1),x_opt(2),'ko','MarkerSize',5,'MarkerFaceColor','b');
xlabel("X");
ylabel("Y");
title("Contour Plot of Objective Function");
legend('Contour Plot','x+y>=1','Optimum Point','Location','northwest');
grid on;
hold off;
```



6. Maximize z=$x + y$ subject to $x^2 + y^2 \leq 25$; $x + y \geq 5$; $x, y \geq 0$

```
clc;
x=0:0.1:1;
y=0:0.1:1;
fun=@(x) x(1)+x(2);
nonlcon=@(x) deal([],x(1)^2+x(2)^2-1);
x0=[0.5,0.5];
lb=[0,0];
[x_opt,fval]=fmincon(fun,x0,[],[],[],[],lb,[],nonlcon);
```

```
disp('Optimal Solution');
disp(['x = ',num2str(x_opt(1)),', y = ',num2str(x_opt(2))]);
disp(['Minimum value of x+y at optimum : ',num2str(fval)]);
[X,Y]=meshgrid(x,y);
Z=X+Y;
contour(X,Y,Z,20,'LineWidth',1.5);
hold on;
theta=linspace(0,2*pi,100);
r=1;
x=r*cos(theta);
y=r*sin(theta);
plot(x,y,'r');
plot(x_opt(1),x_opt(2),'ko','MarkerSize',5,'MarkerFaceColor','b');
xlabel("X");
ylabel("Y");
title("Contour Plot of Objective Function");
legend('Contour Plot','x^2+y^2=1','Optimum point','location','northwest');
axis equal;
grid on;
hold off;
```



7. Minimize z= $x + y$ subject to $x^2 + y^2 \leq 25$; $x + y \geq 5$; $x, y \geq 0$.

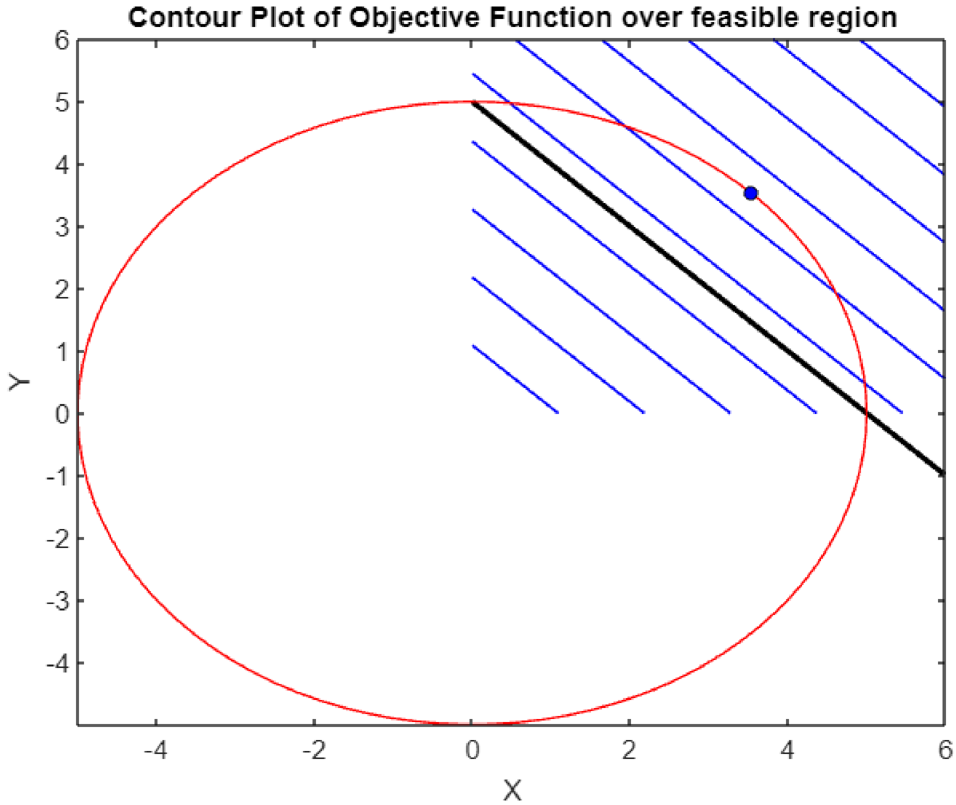
```
clc;
fun=@(x)-(x(1)+x(2));
nonlcon=@(x) deal(x(1)^2+x(2)^2-25,[]);
x0=[0,0];
A=[-1,-1];
b=-5;
lb=[0,0];
[x_opt,fval]=fmincon(fun,x0,A,b,[],[],lb,[],nonlcon);
```

```
fval=-fval;
disp("Optimal solution : ");
```

```
disp(['x = ',num2str(x_opt(1)), ', y = ',num2str(x_opt(2))]);
```

```
disp(['Maximum value of x+y at optimum : ',num2str(fval)]);
```

```
x=linspace(0,6,100);
y=linspace(0,6,100);
[X,Y]=meshgrid(x,y);
Z=X+Y;
contour(X,Y,Z,10,'b','LineWidth',1);
hold on;
plot(x,5-x,'black','LineWidth',2);
theta=linspace(0,2*pi,100);
r=5;
x=r*cos(theta);
y=r*sin(theta);
plot(x,y,'r');
plot(x_opt(1),x_opt(2),'ko','MarkerSize',5,'MarkerFaceColor','b');
xlabel("X");
ylabel("Y");
title('Contour Plot of Objective Function over feasible region');
```



8. Minimize z=$x^2 + y^2$ subject to $x^2 + y^2 \quad \leq 25$; $x - y \geq 1$; $x \leq 3$; $x, y \geq 0$.

```
clc;
clf;
clear all;
```

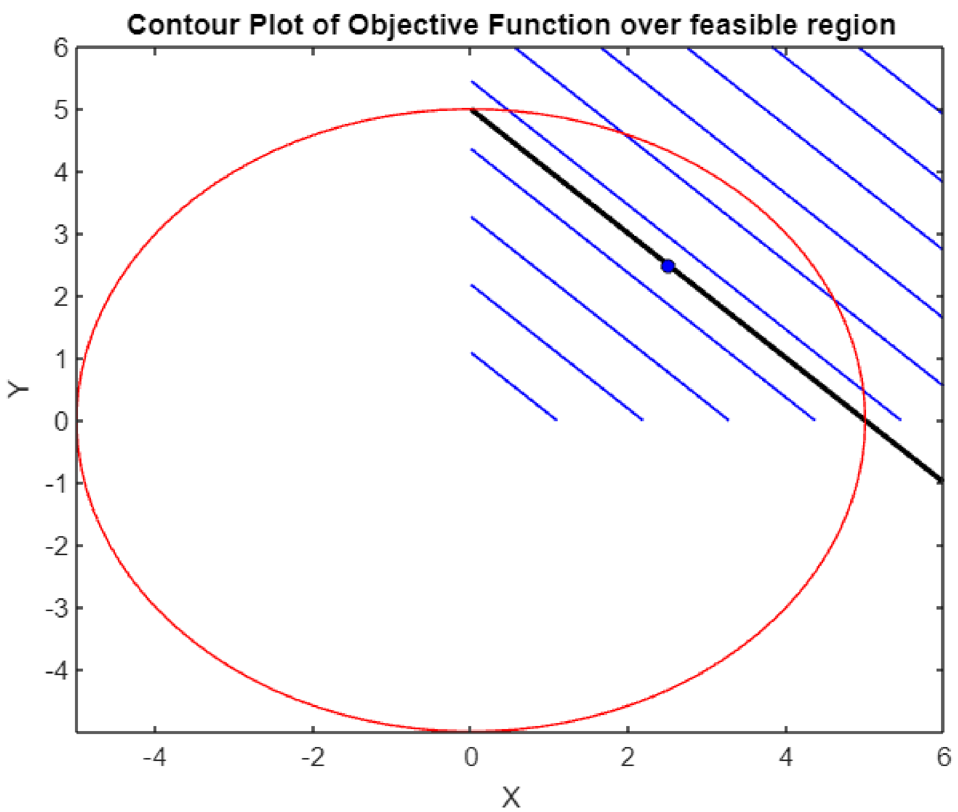
```
fun=@(x) x(1)+x(2);
nonlcon=@(x) deal(x(1)^2+x(2)^2-25,[]);
x0=[0,0];
A=[-1,-1];
b=-5;
lb=[0,0];
[x_opt,fval]=fmincon(fun,x0,A,b,[],[],lb,[],nonlcon);
```

```
fval=-fval;
disp("Optimal solution : ");
```

```
disp(['x = ',num2str(x_opt(1)), ', y = ',num2str(x_opt(2))]);
```

```
disp(['Maximum value of x+y at optimum : ',num2str(fval)]);
```

```
x=linspace(0,6,100);
y=linspace(0,6,100);
[X,Y]=meshgrid(x,y);
Z=X+Y;
contour(X,Y,Z,10,'b','LineWidth',1);
hold on;
plot(x,5-x,'black','LineWidth',2);
theta=linspace(0,2*pi,100);
r=5;
x=r*cos(theta);
y=r*sin(theta);
plot(x,y,'r');
plot(x_opt(1),x_opt(2),'ko','MarkerSize',5,'MarkerFaceColor','b');
xlabel("X");
ylabel("Y");
title('Contour Plot of Objective Function over feasible region');
```



Practice Questions:

1. *Maximize x+y subject to y ≤ 5, x+y ≤10; x,y ≥0*

2. *Maximize 3y subject to x ≤3; -x+y ≤ 4; x+y ≤6 and x, y≥0*

3. *Maximize x+y subject to x ≤ 3, y ≤ 7 and x, y ≥0*

4. *Maximize and minimize* $x^2 + y^2$ *in the first quadrant, subject to x ≤ 3, y ≤ 7 and x, y ≥0*